SMART ELEVATOR HACKATHON 2026 — HARDWARE TRACK

Intelligent Sensor Fusion Module

for Elevator Safety Monitoring

One compact edge device. Five live sensor feeds. A single real-time health score for every elevator subsystem.

---

1. Problem Statement

Elevator subsystems — motor, traction, braking, and doors — largely operate and fail independently, with no unified sensing platform watching them together. As a result, faults such as excess vibration, overheating, or electrical overload often go undetected until a subsystem fails outright, leading to costly downtime, emergency call-outs, and disrupted buildings. Most in-service elevators were never built with sensors, so retrofitting monitoring typically means rewiring the shaft.

- No unified view: each subsystem runs on its own logic, with nobody watching them together.
- Faults surface late: problems build up quietly until a subsystem fails outright.
- Downtime is expensive: reactive repairs mean longer outages and emergency call-outs.
- Retrofitting is hard: adding monitoring to existing elevators usually means rewiring.

2. Objective

Develop a compact edge hardware platform that integrates five sensor types into a single module, giving a real-time health score for every elevator subsystem.

- Unify — pull vibration, temperature, current, position, and door state into one device.
- Digitize — sample every subsystem continuously instead of relying on manual inspection.
- Predict — flag abnormal readings before they become breakdowns, using threshold and trend logic.
- Visualize — surface one live health score and alert feed a technician can act on immediately.

3. System Architecture

Five sensors feed into a central ESP32 Fusion Module, which performs edge processing (threshold checks and rolling averages, sampled every 200 ms) before transmitting data over WiFi (MQTT or HTTP) to a live dashboard.

Data flow: Sensors → ESP32 Fusion Module → Edge Processing → WiFi → Live Dashboard

Hardware Components

Sensor	Component	Purpose
Vibration	MPU6050	Detect abnormal shaking in the cabin or motor housing
Temperature	DS18B20	Watch for motor and brake overheating
Current	ACS712	Catch motor overload and electrical faults early

Position	Rotary encoder	Track cabin position and speed in real time
Door	Magnetic reed switch	Confirm door open / closed state on every cycle

4. Data Pipeline

- Sample — ESP32 reads all five sensors every 200 ms over analog, digital, and I2C pins.
- Filter — edge logic applies threshold checks and a rolling average to cut sensor noise.
- Transmit — readings are pushed over WiFi via MQTT or HTTP to the dashboard backend.
- Score — the backend combines subsystem readings into one live health score, 0–100%.
- Alert — any threshold breach raises an instant alert naming the responsible subsystem.

5. Demo Dashboard

The live dashboard shows a single system health score alongside subsystem status, a predictive analytics trend, and an active alerts feed — the primary artifact judges will see on stage.

- System Health: 92% — GOOD, no active alerts (example reading).
- Subsystem status: Motor, Traction, Brakes — normal; Doors — flagged.
- Predictive analytics: vibration trend rising over a 6-hour window.
- Active alerts: door subsystem approaching threshold, forecast in ~4 hours — inspection recommended.

6. Impact

Benefit	Why it matters
Fewer breakdowns	Early threshold alerts catch faults before they strand a cabin mid-shaft.
Lower service cost	Technicians act on real data instead of routine, calendar-based inspections.
Faster retrofits	One compact module bolts onto existing elevators — no rewiring the shaft.
Fleet-ready	The same edge module and dashboard scale from one cabin to an entire building.

7. 48-Hour Build Plan

Phase	Activities
H0–H6	Wire sensors to ESP32, confirm each reads clean values on serial monitor.
H6–H16	Write firmware: sampling loop, threshold logic, WiFi transmit.
H16–H30	Build dashboard: live health score, trend chart, alert feed.
H30–H40	Integration testing: trigger real faults, tune thresholds.
H40–H48	Polish demo, rehearse the live-shake alert trigger.

Team Name: [your team here]